

Featured engineering

In this step, I created new variables from the existing insurance data, including claim frequency per customer, average claim severity, and policy claim summary. These features were generated using SQL transformations and stored as SQL views within the database, making them easily accessible for analysis, reporting, and future modelling. These are the queries used to create these SQL views:

Policy Claim Summary:

- ```
CREATE VIEW policy_claim_summary AS
SELECT
 policyid,
 COUNT(claimsid) AS number_of_claims,
 SUM(Amount) AS total_claim_amount,
 AVG(Amount) AS avg_claim_amount
FROM claims
GROUP BY policyid
```

Average Claim severity:

- ```
CREATE VIEW average_claim_severity AS
SELECT
    AVG(Amount) AS avg_claim_severity
FROM claims
```

Customer Claim frequency

- ```
CREATE VIEW customer_claim_frequency AS
SELECT
 p.customerid,
 COUNT(c.claimsid) AS claim_frequency
FROM policy p
INNER JOIN claims c
 ON p.policyid = c.policyid
GROUP BY p.customerid
```

Customer Claim Summary

- ```
CREATE VIEW customer_claim_summary AS
SELECT
    p.customerid,
    COUNT(c.claimsid) AS number_of_claims,
    SUM(c.amount) AS total_claim_amount,
    AVG(c.amount) AS avg_claim_amount
FROM policy p
INNER JOIN claims c
    ON p.policyid = c.policyid
GROUP BY p.customerid
```